

Introduction to the Virtual Collection of Discrete Simulation of Fluid Dynamics

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THIS virtual collection contains peer-reviewed computational and theoretical research papers based on presentations given at the 32nd Discrete Simulation of Fluid Dynamics (DSFD) conference, which was organized by the University of New Mexico and the Los Alamos National Laboratory and held at the University of New Mexico, Albuquerque, NM, July 17–21, 2023. The DSFD series of conferences originated with the 1986 Los Alamos conference organized by Gary D. Doolen. Since then, DSFD conferences have emerged as the premier forum for researchers in the field, and many exciting discoveries in lattice models of fluid dynamics have been first announced at DSFD conferences. Topics emphasized at these meetings include lattice Boltzmann method (LBM) schemes, dissipative particle dynamics, smoothed-particle hydrodynamics, direct simulation Monte Carlo, molecular dynamics, quantum Monte Carlo methods, multiparticle collision dynamics, and hybrid methods. The conference is held annually, but due to COVID restrictions, the 2020 edition to be held in Viterbo, Italy, had to be organized online. In 2021, the organizers were hopeful to finally hold the conference in person in Viterbo, but conditions still were not suitable for an in-person conference. The 31st DSFD conference in 2022 was then scheduled to be held in person in Suzhou, China. However, a resurgence of COVID caused severe lockdown in the area before the conference, leaving the only option for a hybrid conference, and most international attendees only participated online. So after the COVID lockdowns, the 32nd DSFD conference was the first to be fully in person again after the pandemic. It was a great pleasure to see many familiar faces again and to meet new researchers in the field in person for the first time.

Of the numerical methods that have been at the focus of the DSFD conferences, lattice Boltzmann in particular has grown enormously in popularity in the nearly 4 decades since the seminal paper by Frisch, Hasslacher, and Pomeau [1]. Not overly surprisingly, lattice Boltzmann contributions tend to dominate the presentations at this conference.

The papers in the collection cover the following topics:

- 1) Contributions to LBM
- 2) LBM simulations
- 3) Machine learning (ML) augmentations to LBM
- 4) Beyond LBM with classical neutral fluids: Magneto-hydrodynamics (MHD) and quantum nanofluids extensions
- 5) Other particle methods (molecular dynamics, MD) and extensions (droplet representations)

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Thus, Yan and Shan [2] discuss a spectral multiple-relaxation-time (SMRT) lattice Boltzmann (LB) model and show that it is equivalent to Grad's eigensystem and represents a fast discretization of the moment method. Quintana et al. [3] propose a forcing method to capture multicomponent phase-separating systems using the incompressible LBM. Li et al. [4] discuss a conservative phase-field LBM model based on the Allen–Cahn equation for simulating immiscible two-phase flows with large density and viscosity ratios. Klass et al. [5] compared the numerical accuracy and cost of perfectly matching layers and characteristic boundary conditions for compressible LBM. Hettel et al. [6] demonstrate LBM for calculations of multiple free jets in different configurations. Bukreev et al. [7] perform a benchmark simulation of a reacting micromixer considering reacting multicomponent flows with LBM. Montessori et al. [8] develop a thread-safe LBM numerical approach on GPU-based architectures and deploy it to simulate axisymmetric turbulent jets of a fluid evolving in a quiescent, immiscible environment. Ortali et al. [9] propose lattice-equivariant neural networks, designed to satisfy local symmetries of a lattice structure to enhance LBM calculations. Dellar [10] discusses LBM extensions for MHD. Succi et al. [11] propose a Keldysh–lattice Boltzmann approach to quantum nanofluids. Van Swol et al. [12] perform MD simulations of fluid droplets impacting surfaces. Finally, Durve et al. [13] demonstrate autoencoders for the representation of liquid droplet surfaces with a small number of parameters. This virtual collection aims to highlight the newest developments in particle methods for fluid dynamics and further draw attention to these methods in the aerospace community.

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